👑 CODERKING STUDIO

Official Technical Roadmaps & Architecture Guides

HIGH-SCALE FINANCIAL ENGINEERING

Zerodha Architecture: Scaling 1.5Cr Trades on ₹0 Cloud Bill

How 35 engineers built India's largest stock brokerage with bare-metal hardware and open-source software.

🏢 1. Bare Metal Colocation vs Cloud Bill Shock

Most modern startups burn millions on AWS/GCP managed instances and egress network taxes.

Zerodha's Strategy: Leased bare-metal servers in dedicated Mumbai data centers with enterprise SSDs and ultra-low latency direct exchange connectivity.

- **Cost Savings:** Saved hundreds of crores in recurring cloud markup.
- **Performance:** Deterministic bare-metal CPU performance with zero noisy-neighbor virtualization jitter.

🐘 2. Vanilla PostgreSQL Scaling

Resisted the hype of complex distributed NoSQL databases that introduce network splits and consensus lag.

- **Table Partitioning:** Trades and orders partitioned cleanly by trading day.
- **Read Replicas:** Real-time streaming replication offloads portfolio read queries.
- **Fast SSDs + Huge RAM:** Active order book stays 100% in memory cache.

⚡ 3. High-Concurrency Go Microservices

Kite trading web and mobile APIs are powered by lightweight Go services:

```
Client Request -> HAProxy Load Balanc
er -> Go Goroutines -> Redis Order St
ream -> OMS Engine
```

Each Goroutine requires only 2KB of RAM, enabling 100,000+ simultaneous market tick streams per node.

💡 4. The FOSS-First Engineering Playbook

Key architectural lessons from Zerodha CTO Kailash Nadh:

- **Keep Dependencies Minimal:** Fewer third-party SaaS vendors means fewer failure points.
- **Small Elite Team:** 35 engineers communicate with zero managerial bureaucracy.
- **Build Tools In-House:** Custom order matching and risk management engines tailored for Indian markets.